Polarization, LASER
&
Fibre Optics

Polarization

- The transverse nature of waves leads to the characteristic phenomenon called **polarization**. Longitudinal waves do not exhibit the characteristic of polarization.

- In a transverse wave, if the directions of all the vibrations at all the points are restricted to one **particular plane**, then the wave is called **polarized**, more specific plane polarized. A plane-polarized wave is the simplest transverse wave, also termed a linearly polarized.

- According to the electromagnetic theory, light consists of electric and magnetic vectors vibrating continuously with time in a plane, transverse to the direction of propagation of light and each other. However, **in explaining polarization only the vibrations of the electric vector are considered**. It does not mean that magnetic field vectors are absent, they are present. But for drawing simplicity they are not shown in the diagram.

Polarization – Un-polarized Light

- The light having vibrations along all possible directions perpendicular to the direction of propagation of light, is called an **unpolarized light**. It is symmetrical in its direction of propagation.

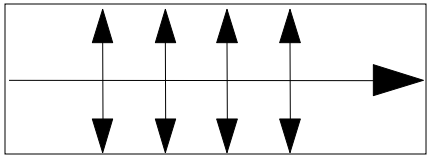


Unpolarized light

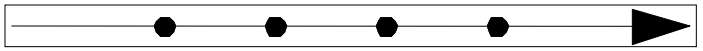
- It can be considered to consist of an infinite number of waves each having its own vibration. Therefore, for any position of the crystal there will be one vibration parallel to its axes, so when such a light is passed through a single tourmaline crystal and is rotated no change in the intensity of the emergent light is observed.
- Since unpolarized light has vibrations along all possible directions, at right angles to the directions of propagation of light, it is represented by a star.

Polarization – Polarized Light

- The light having vibration only along a single plane perpendicular to the direction of propagation of light is called a **polarized light**. Its vibrations are one sided, therefore it is dissymmetrical about the direction of propagation of light. When polarized light is passed through a single rotating crystal, a change in the intensity of emergent light is observed.
- According to the electromagnetic theory, a plane polarized light can be defined as follows :



(a) Parallel to plane of paper



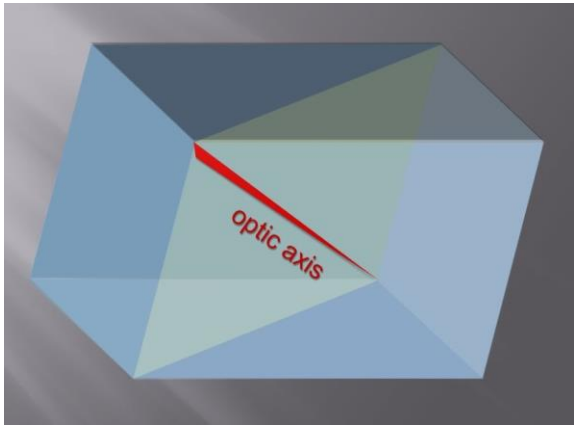
(b)Perpendicular to plane of paper

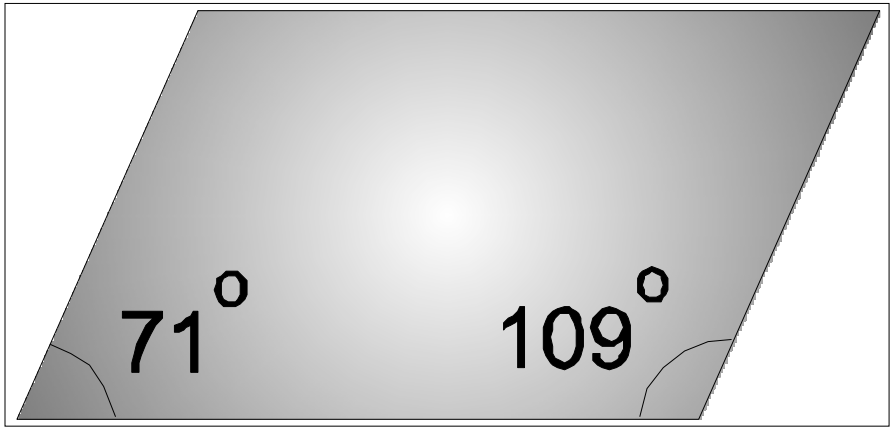
- A light wave is said to be **plane polarized** if the electric vector (E) or magnetic vector (H) at any point is vibrating along the same plane perpendicular to the direction of propagation of the light wave.
- The polarized beam of light has vibrations along a single plane. If they are parallel to the plane of the paper, they are represented by arrows [See Fig. (a)]. If they are perpendicular to the plane of the paper, they are represented by dots on a ray of light. [See Fig. (b)].

Principal Section & Principal Plane

- **Principal Section –** A plane containing **the optic axis** and **perpendicular to the opposite faces** of the crystal is called the principal section of the crystal. The principal section cuts the surfaces of a calcite crystal in a parallelogram with angles 109° and 71°.

- **Principal Plane –** The plane containing **the optic axis,** and **the ordinary ray** is called **principal plane of the ordinary ray**. Similarly, the plane containing **the optic axis,** and **the extraordinary ray** is called **the principal plane of the extraordinary ray**.

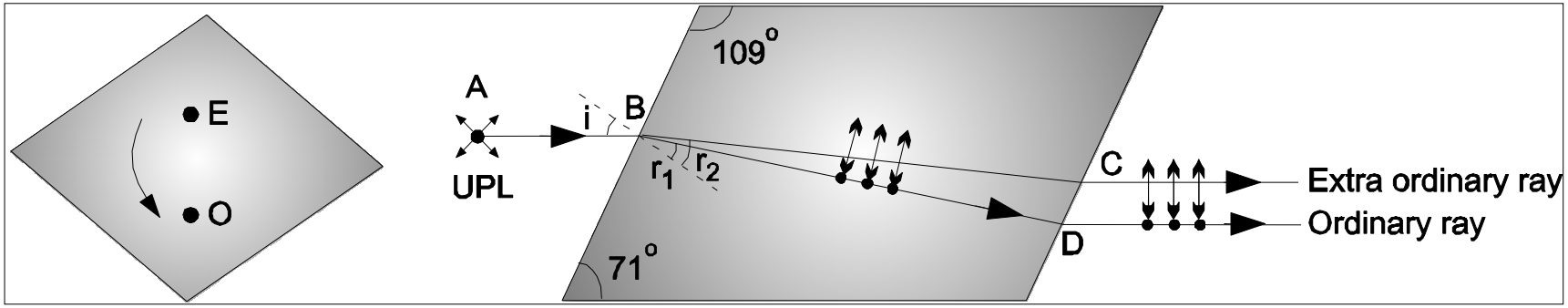




Principal section of calcite crystal

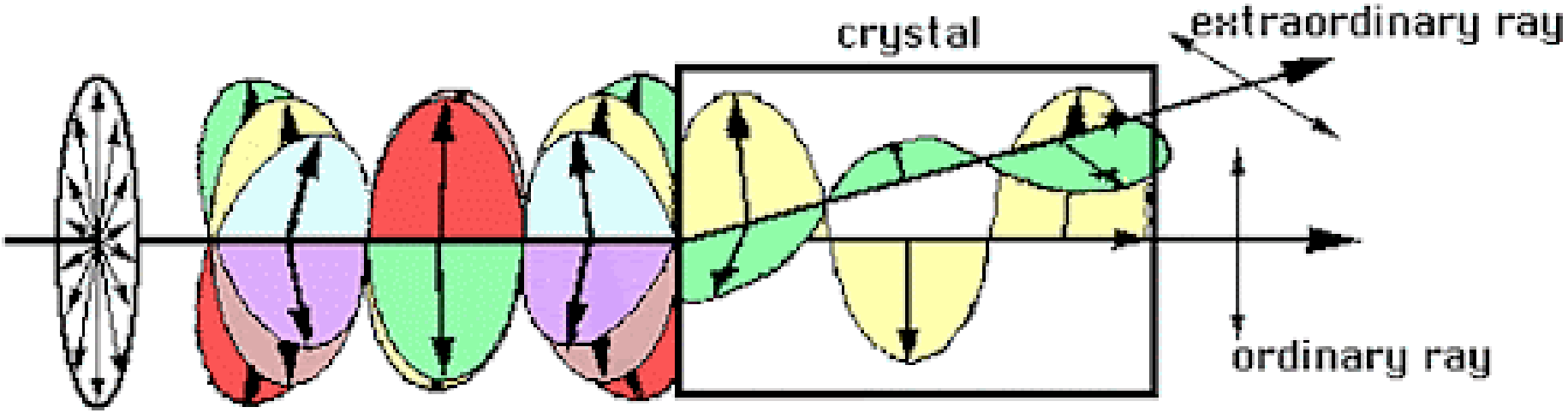
Polarization – Double refraction

- The phenomenon of double refraction was discovered by Erasmus Bartholinus in 1669 during his studies on calcite. When light is incident on a calcite crystal, it is found to produce two refracted rays which are different in properties. The phenomenon of causing **two refracted rays** by a crystal is called **birefringence** or **double refraction**. The crystals are said to be **birefringent**.



- All anisotropic materials exhibit double refraction. The two rays formed in double refraction are linearly polarized in mutually perpendicular directions. One of the rays **obeys Snell's law** of refraction and hence is called **an ordinary ray or O-ray**.
- The other ray **does not obey Snell's law** and is called an **extraordinary ray or E-ray.** If one of the rays is eliminated, the light transmitted by the crystal will be linearly polarized.
- When a ray of light AB is incident on the calcite crystal making an angle of incidence i, it is refracted along two paths inside the crystal :
  (i) along BC making an angle of refraction r_2,
  (ii) along BD making an angle of refraction r_1.

These two rays emerge out along DO and CE which are parallel as the crystal faces are parallel.

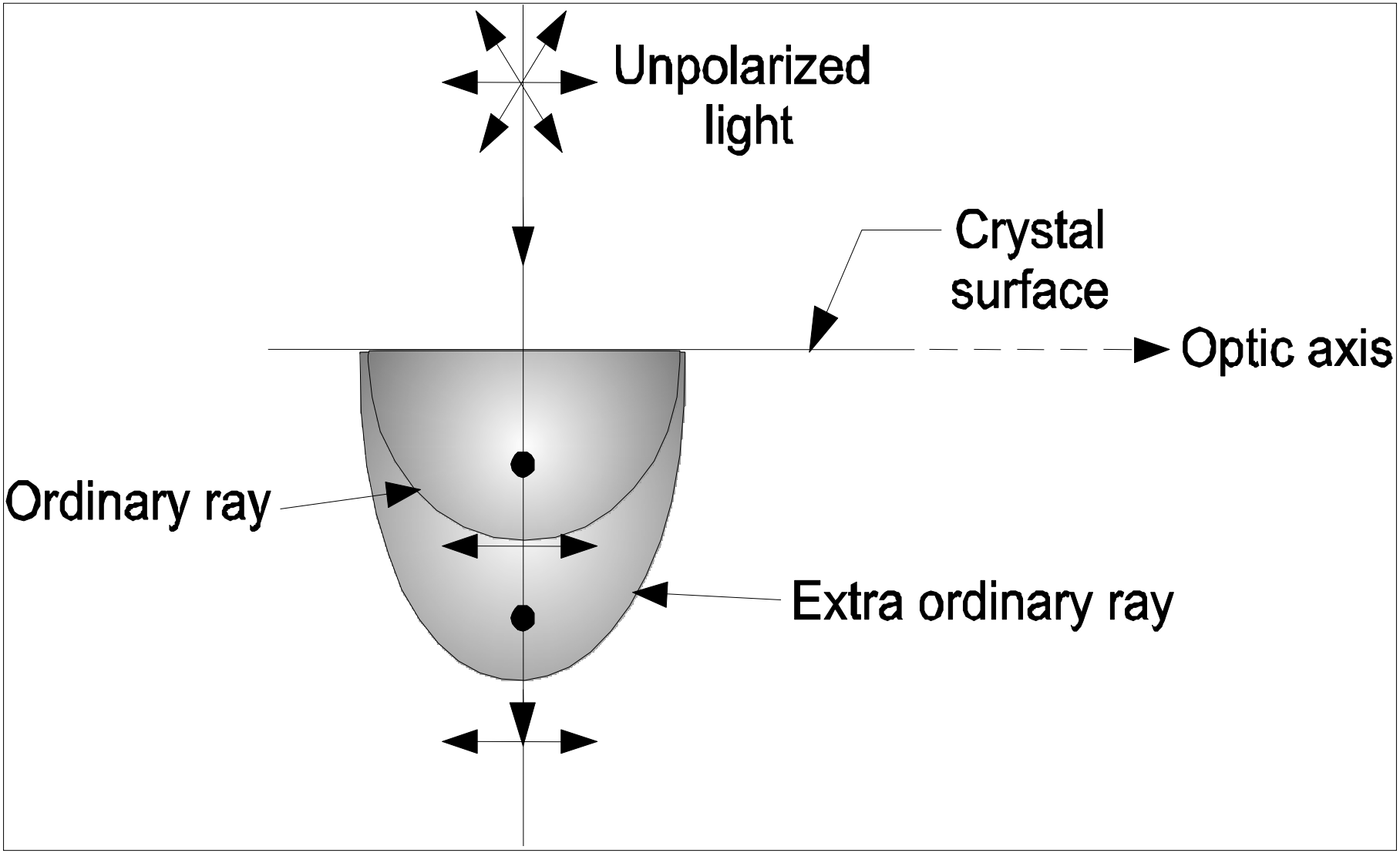


Polarization – Huygen's Theory

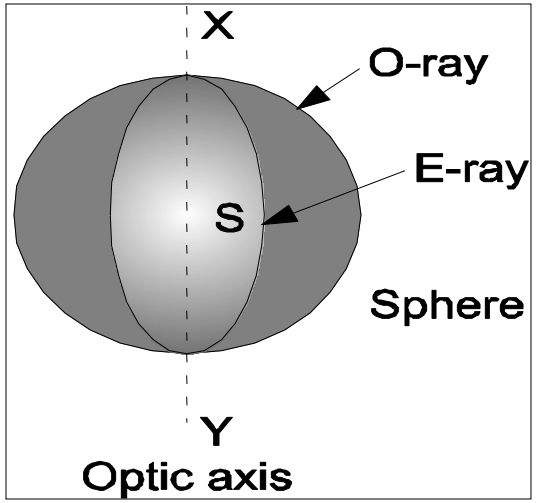
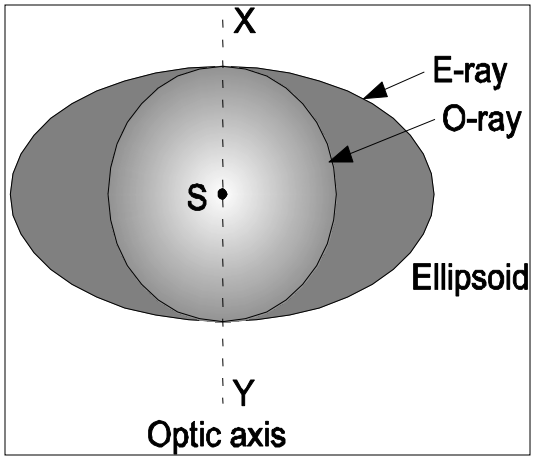
Huygen explained the phenomenon of double refraction based on the principle of secondary wavelets. He assumed :

1. When a beam of ordinary unpolarized light strikes a doubly refracting crystal, each point on the surface sends out **two wavefronts**, one for ordinary rays and the other for extraordinary rays.
2. The **ordinary ray** travels with the **same speed** v_o in all directions and the crystal has a single refractive index $\mu_o = c/V_0$ for this wave. Thus, the O-ray has a **spherical wavefront**.
3. The **speed of extraordinary rays v_e varies with direction**. So, the refractive index, $\mu_e = c/V_e$ also varies with direction for the E-ray. Therefore, the extraordinary ray develops a wavefront that is **ellipsoidal**.
4. The velocity v_e measured is perpendicular to the optic axis.
5. The velocities of the O-ray and E-ray are the same along the optic axis.
6. When rays are incident along the optic axis, the spherical and ellipsoidal wavefronts touch each other at points of intersection with the optic axis and double refraction does not take place.
7. If $v_o > v_e$ or $\mu_o < \mu_e$, the spherical wavefront lies outside the elliptical wavefront. Such crystals are called **positive crystals**. Examples of positive crystals are quartz, ice, etc.
8. If $v_e > v_o$ or $\mu_e < \mu_o$, the elliptical wavefront lies outside the spherical wavefront. Such crystals are called **negative crystals**. Examples of negative crystals are calcite, tourmaline, etc.

Polarization – Huygen's Theory



Polarization – Positive & Negative Crystals

Positive Crystals	Negative Crystals
For positive crystals, $v_o > v_e$ and $\mu_o < \mu_e$.	For negative crystals, $v_o < v_e$ and $\mu_o > \mu_e$.
The velocity of O-ray is same in all directions.	The velocity of O-ray is same in all directions.
The wavefront of O-ray lies outside the wavefront of E-ray.	The wavefront of O-ray lies inside the wavefront of E-ray.
Examples : Quartz, Ice. 	Examples : Calcite, Tourmaline. 

Polarization – Nicol Prism

Nicol prism is an optical device used for **producing and analyzing plane-polarized light**.

***Principle*:** The Nicol prism is made in such a way that it eliminates one of the refracted rays by total internal reflection i.e. O-ray is eliminated and only E-ray is transmitted through the prism.

***Construction*:** A **calcite crystal** of length three times its breadth is taken. Let ABCD is the principal section of the crystal with ∠ BAD = 71°. The end faces of the crystal are cut in such a way that they make angles 68° and 112° in the principal section instead of 71° and 109°.

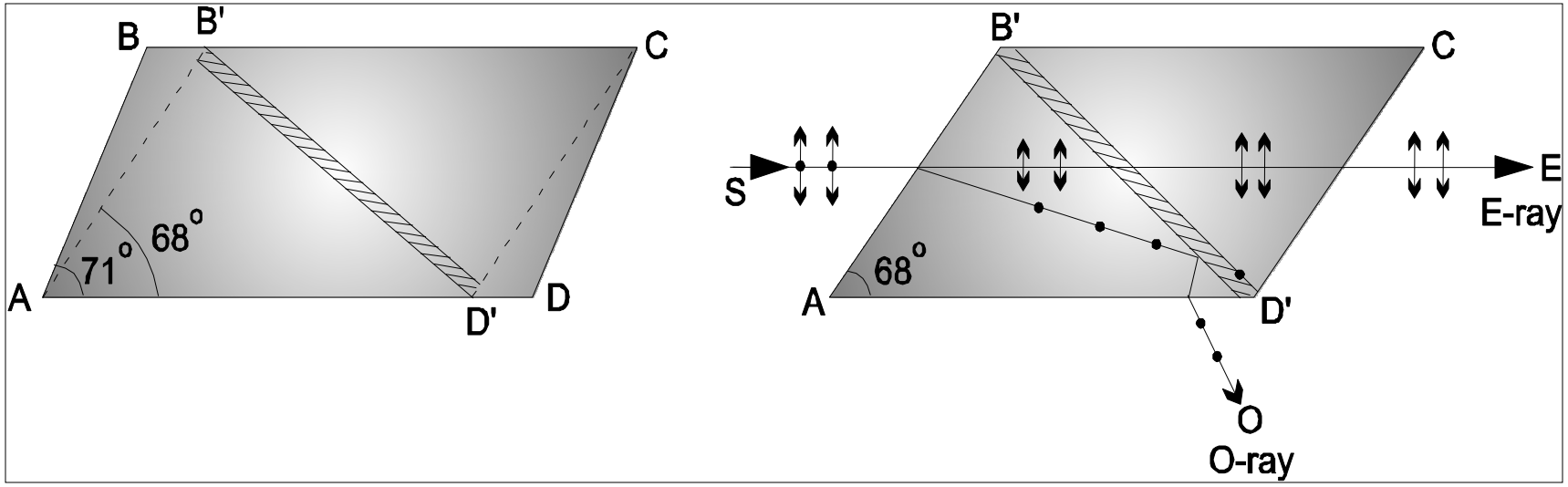
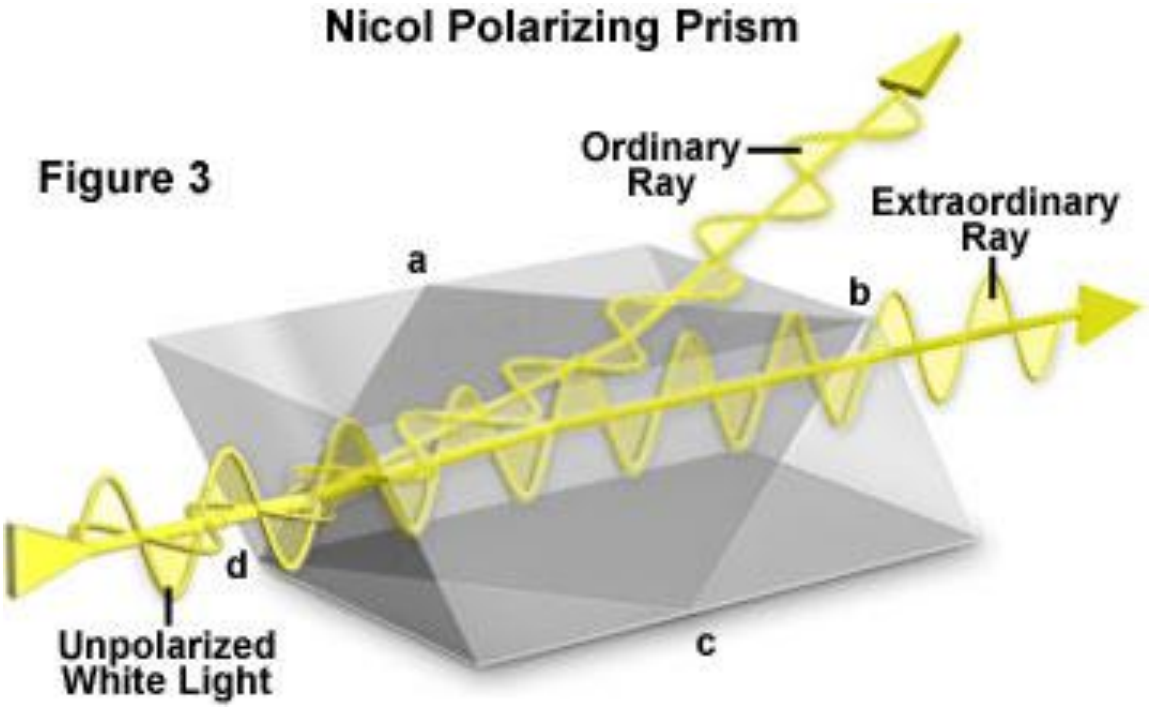


Figure 3


Nicol Polarizing Prism
Ordinary Ray
Extraordinary Ray
a
b
c
d
Unpolarized White Light

Polarization – Nicol Prism

The crystal is then cut into two pieces from one blunt corner to the other along a plane perpendicular to the principal section. The two cut faces are grounded and polished optically flat.

It is then cemented together by Canada balsam of refractive index lies between the refractive indices for the O-ray and E-ray for calcite.

The refractive index of Calcite for O-ray

$$\mu_o = 1.658$$

Refractive index of Canada Balsam

$$\mu_c = 1.55$$ Using sodium light of $\lambda = 5893A^\circ$,

The refractive index of Calcite for E-ray

$$\mu_e = 1.486$$

Canada balsam layer acts as a rarer medium for O-ray and as a denser medium for E-ray. Except for the end faces, the sides of the crystal are blackened.

Working – When a ray of unpolarized light is incident on the prism surface, it splits into O-ray And E-ray.

Both the rays are polarized having vibrations at right angles to each other. When the O-ray passes from a portion of the crystal into the layer of Canada balsam, it passes from a denser medium to a rarer medium.

When the angle of incidence is greater than the critical angle, the O-ray is internally reflected and is not transmitted.

When the E-ray passes from calcite to the Canada Balsam layer, it enters in rarer medium.

Therefore, the E-ray is not Affected and is transmitted through the prism.

Polarization – Nicol Prism



Working : Refractive index for O-ray with respect to Canada balsam is

$$\mu = 1.658/1.55$$

If C is the critical angle, $\mu = 1/\sin C$

$\sin C = 1/\mu = 1.55/1.658$

$\therefore \quad C = 69°$

As the length of the crystal is large, the angle of incidence at Canada balsam surface for the O-ray is greater than the critical angle. Thus, it suffers total internal reflection while E-ray is transmitted which is plane polarized having vibrations in the principal section.

Special Cases :

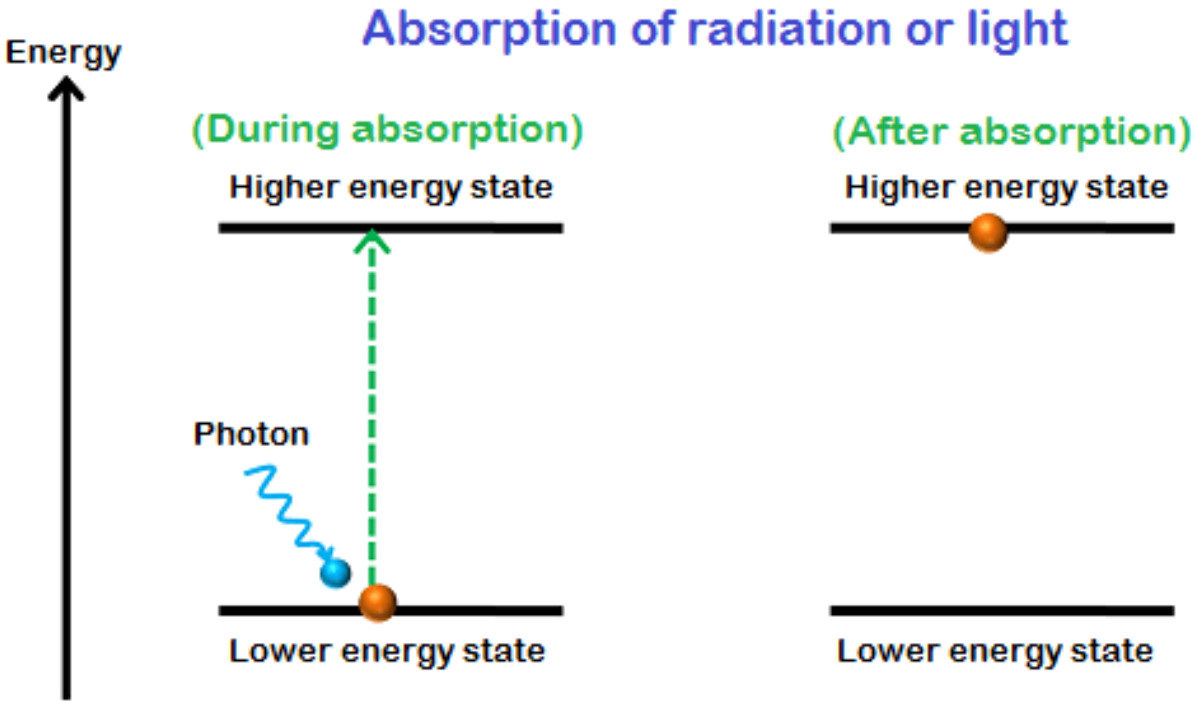
1. If the angle of incidence is less than the critical angle for O-ray, it is not reflected and is transmitted through the prism. In this position, both the O-ray and E-ray are transmitted through the prism.
2. The E-ray also has a limit beyond which it is totally internally reflected by Canada balsam surface. If E-ray travels along the optic axis, its refractive index is the same as that of O-ray i.e. 1.658. But it is 1.486 for all other directions of E-ray. Therefore, depending on the direction of propagation of E-ray, μ_e lies between 1.486 and 1.658. Therefore, for a particular case, μ_e may be more than 1.55 and the angle of incidence will be more than the critical angle. Then E-ray will also be totally internally reflected.

Principle of LASER

- Term laser stands for **'light amplification by stimulated emission of radiation'**. Laser is a light source which is highly coherent i.e., radiation emitted by all the emitters (atoms or molecules) are in the same phase, and monochromatic.

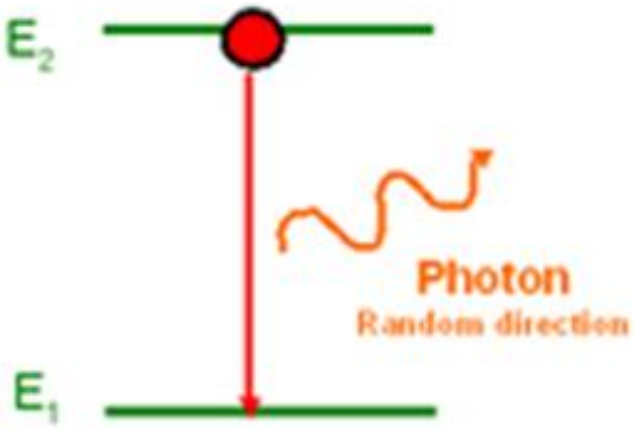
- Due to coherence, a beam of laser light can travel many miles with only a negligible divergence. This makes it different from the conventional light sources which emit many wavelengths with phase and direction widely varying.

Absorption



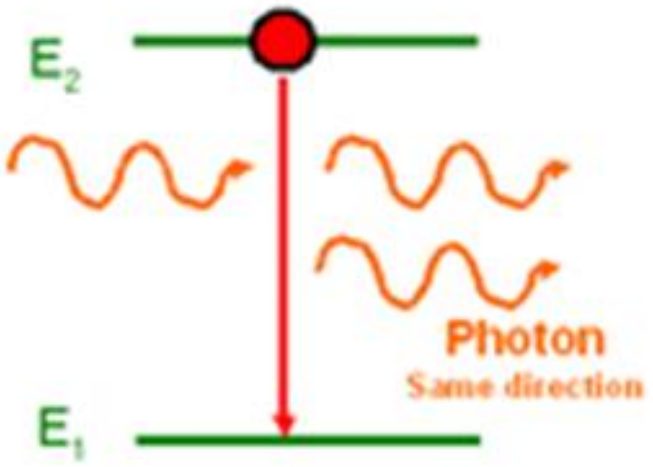
When a photon of energy hv is incident on an atomic system, the atom gets excited from a lower energy E_1 to a higher energy E_2, if the energy of photon equals the difference in the energy levels i.e. hu = $E_2 - E_1$. In this process, the photon gets absorbed, and the atom is said to be excited.

Spontaneous emission



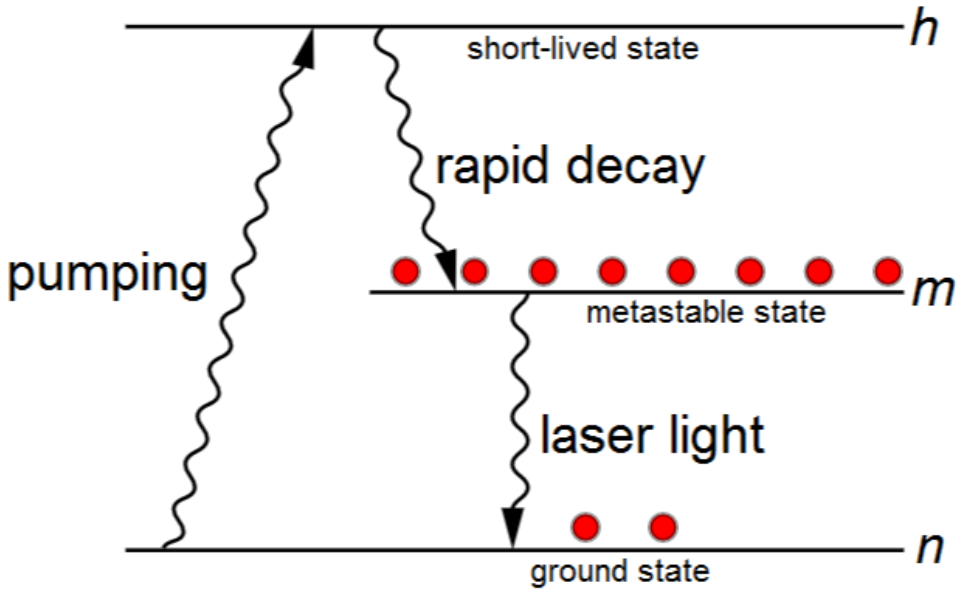
- When a photon of energy E = hu is incident on an atom, the atom absorbs the energy to rise to an excited state. The difference of the energy levels should be equal to the absorbed energy ($E_2 - E_1$ = hu). Though the electron stays in the ground state for an infinite amount of time (being a stable state), but it remains in the excited state for a short time called the **'lifetime'** of the excited state (» 10^{-8} sec). Energy levels for which the lifetime is greater than 10^{-8} sec are called as **'metastable states'**.
- The electron which is in an excited state E_2, spontaneously decays back to the lower energy level and radiates an energy equal to $E_2 - E_1$ in the form of photon of energy hu. Such an emission which is random in behaviour, depending only on the type of atom and the type of transition is known as **'spontaneous emission'.** The emitted radiations are not coherent and have a large range of wavelengths.

Stimulated emission



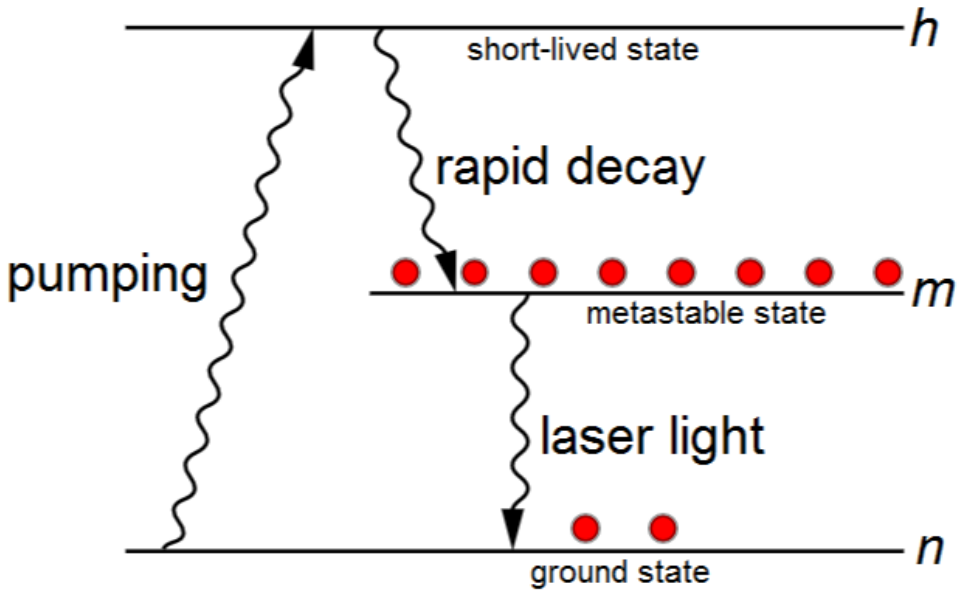
- Consider the Fig. where the electrons are initially in the excited energy level and emission is stimulated before the spontaneous emission occurs. The excited atom is stimulated by a photon of exactly the same energy as the photon to be emitted. In such a case, two photons are emitted, one by the stimulated emission and the other stimulating photon. Both the photons travel in the same direction, have the same frequency and are in phase i.e. they are coherent. The emission of two photons with an input of only one photon implies amplification. The occurrence of spontaneous emission is directly proportional to the number of atoms in the specified energy level, whereas in stimulated emission, the rate of occurrence is proportional not only to the number of atoms in the excited state but also to the number of incident stimulating photons.

Population inversion



- The process of getting a large percentage of atoms into an excited state is called as **'population inversion'.** If a large number of atoms can be excited to upper energy levels, then the probability of stimulated emission and hence light amplification becomes greater. The states of the system, in which the population of the higher energy state is more than the population of the lower energy state, are called as **'negative temperature states'.**

Metastable State



- The electron in an excited state has certain probability to decay or jump to a lower energy level. Generally, these probabilities are such that the jump occurs within 10^{-8} sec of excitation. However, there are some excited states, called **'metastable states'**, which have a very low probability of decay i.e. electrons stay for longer time. Electrons may stay in the metastable excited states for seconds, minutes or even hours. In stimulated emission, the electrons must remain in excited level and wait for stimulating photon. Therefore, the active medium must have a metastable state. The population inversion can be obtained by using metastable states as the electrons rest in metastable state for long time.

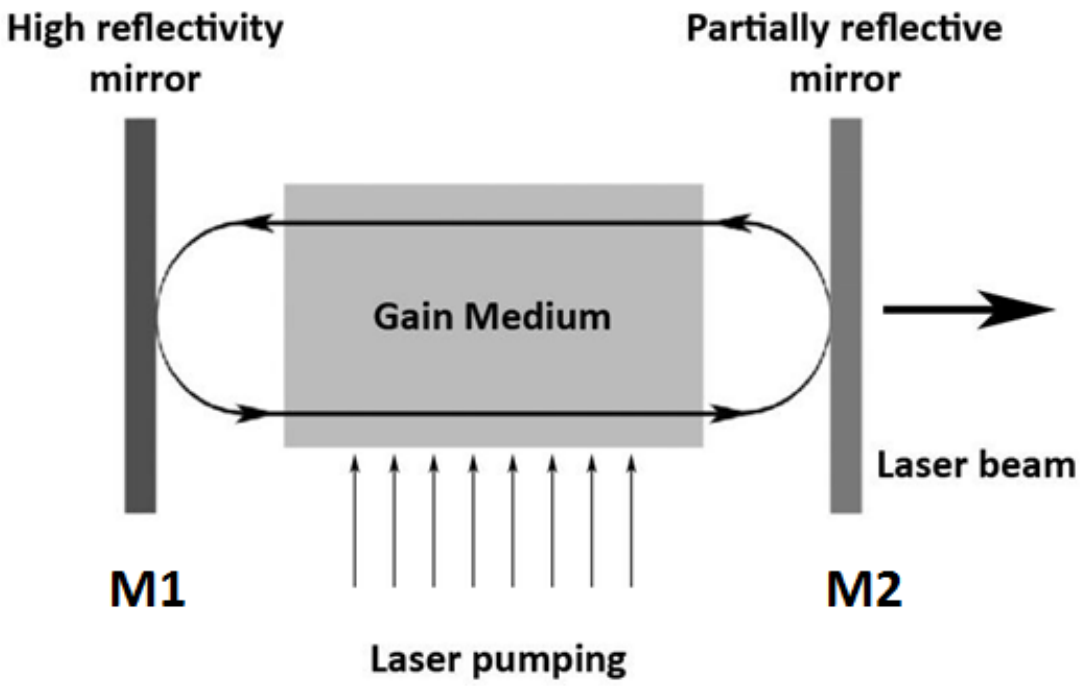
Active Medium, Pumping

- A medium in which the population inversion takes place is called the **active medium**. The active medium is responsible for the light amplification and hence LASER. The active medium may be a solid, liquid or gas and accordingly the lasers are classified as solid state or gas lasers. Out of the total active medium, only small number of atoms are responsible for lasing action and remaining atoms help only in hosting active atoms or in population inversion. The atoms which particulate in stimulated emission are called **active centres**.

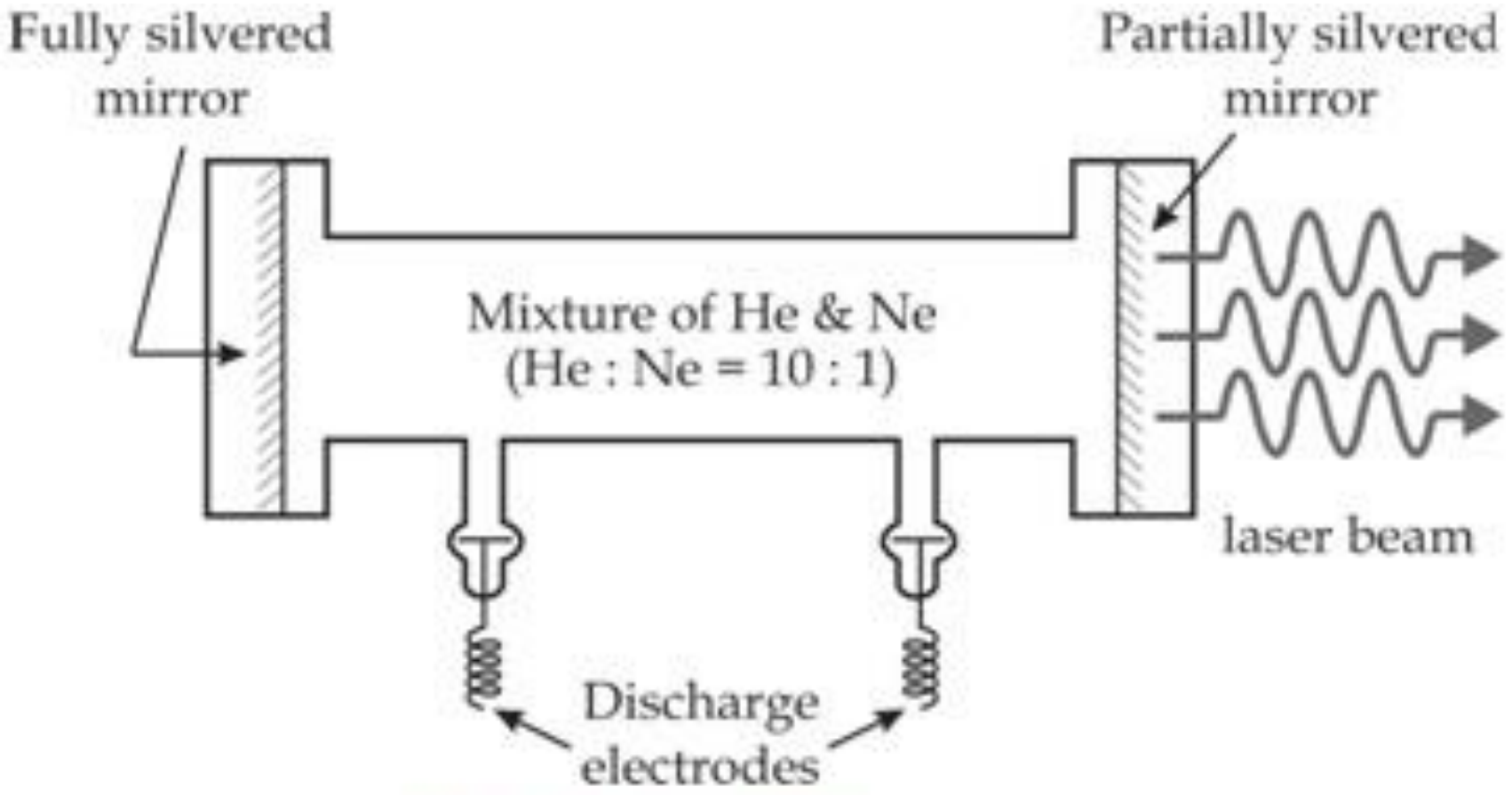
- The method of raising atoms from lower energy levels to higher energy levels is called as **'pumping'**. The pumping is used for achieving population inversion which is necessary for optical amplification to take place. There are several methods for pumping atoms. They are as follows : optical pumping, electrical pumping.

- Spontaneous emission naturally tends to empty the upper level, so this level has to be emptied faster by stimulated emission. It has been proved that stimulated emission is much more likely to happen if the medium used is flooded with light (i.e. with a large number of photons). A good way to do this is to confine the photons in an **optical cavity.**

Resonant Cavity



A cavity can be constructed using mirrors such that the light rays return to their original position after travelling through the cavity for a certain number of times. Such cavities are known as **resonant cavities**. Cavity formed by two parallel mirrors M_1 and M_2. One of the mirrors is completely silvered (M_1) and the other is partially silvered (M_2). The laser beam emerges from the resonant cavity through the partially silvered mirror M_2. The active system is placed in the resonant cavity, the photon emitted will keep on reflecting and forth within the cavity. The light which is incident parallel to the axis of optical cavity will only leak out as a laser. That is why, laser is highly directional.


Fully silvered
mirror
Partially silvered
mirror
Mixture of He & Ne
(He : Ne = 10 : 1)
laser beam
Discharge
electrodes

He-Ne LASER

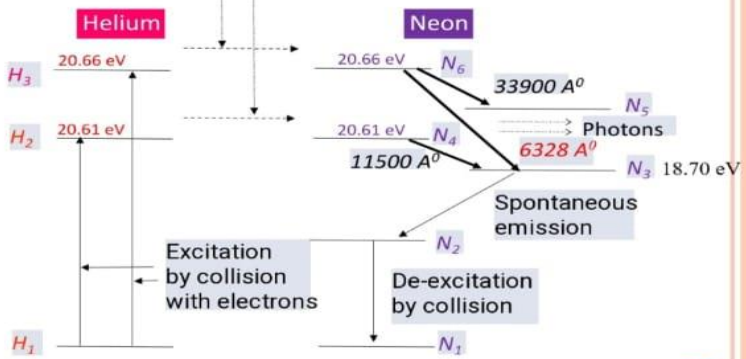
Length of quartz tube ≈ 50 cm & Diameter of quartz tube ≈ 1 cm.

Ratio of He-Ne in the quartz tube is 10:1.

Pumping is done by electric discharge.

Energy Level Diagram:

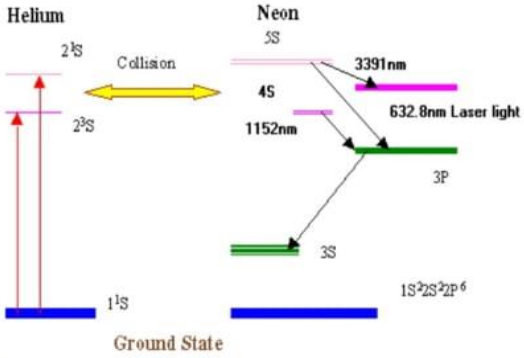
Energy transfer through atomic collision



He-Ne Laser is a Continuous laser.

Working of He-Ne Laser

$$A^* + B \longrightarrow B^* + A$$



Electric discharge is passed through the gas.

Electrons are accelerated, collide with He and He atoms and excite them to higher energy levels.

e^-s – collide with He + Ne
85% He excited more readily than Ne

$$He + e^- \longrightarrow He^*$$

$$He^* + Ne \longrightarrow Ne^* + He$$

Helium-Neon laser

This is a **'continuous laser'** unlike the ruby laser. The first gas laser to be operated successfully was the He - Ne laser. In solid-state lasers, pumping is usually done by using a flash lamp or a continuous high-power lamp. Such a technique is efficient if the laser system has broad absorption bands. In gas lasers, as the atoms are characterized by sharp energy levels, an electrical discharge is generally used to pump the atoms.

Construction: It consists of a quartz tube with a diameter of about 2-8 mm and a length of 10-100 cm. It is filled with helium and neon. The pressure of helium is approximately 10 times that of neon. The neon atoms provide energy states for the transitions. In contrast, helium provides a mechanism for efficiently exciting neon atoms to upper metastable states i.e. helium serves merely as an energy transfer agent. At one end of the tube is a total reflector while at the other is a partial reflector. The gas is excited using an electric discharge.

Helium-Neon laser

Principle :

In He-Ne laser, population inversion is produced through inelastic collisions between excited He atoms and Ne atoms in the ground state. The process can be expressed as

$$He^* + Ne \rightarrow He + Ne^* \quad (* \text{ shows an excited state})$$

This is possible, because the levels Ne_4 and Ne_6 of neon atoms have almost the same energy as the levels He_2 and He_3 of helium atoms as shown in the energy level diagram.

Working :

(a) Electric Discharge and Excitation of Helium :

When an electrical discharge is passed through the gas, the electrons which are accelerated down the tube collide with helium and neon atoms and excite them to higher energy levels. The helium atoms tend to accumulate at the levels He_2 and He_3 due to their long lifetimes of » 10^{-4} secs and 10^{-6} secs respectively.

(b) Transfer of Energy from Helium to Neon and Pumping :

As the levels Ne_4 and Ne_6 of neon atoms have almost the same energy as He_2 and He_3, excited helium atoms colliding with neon atoms in the ground state can excite the neon atoms to Ne_4 and Ne_6 states. As the pressure of helium is ten times that of neon, the levels Ne_4 and Ne_6 of neon are selectively populated as compared to other levels of neon.

i.e. $He^* + Ne \rightarrow He + Ne^*$ (* indicates excited state)

Helium-Neon Laser

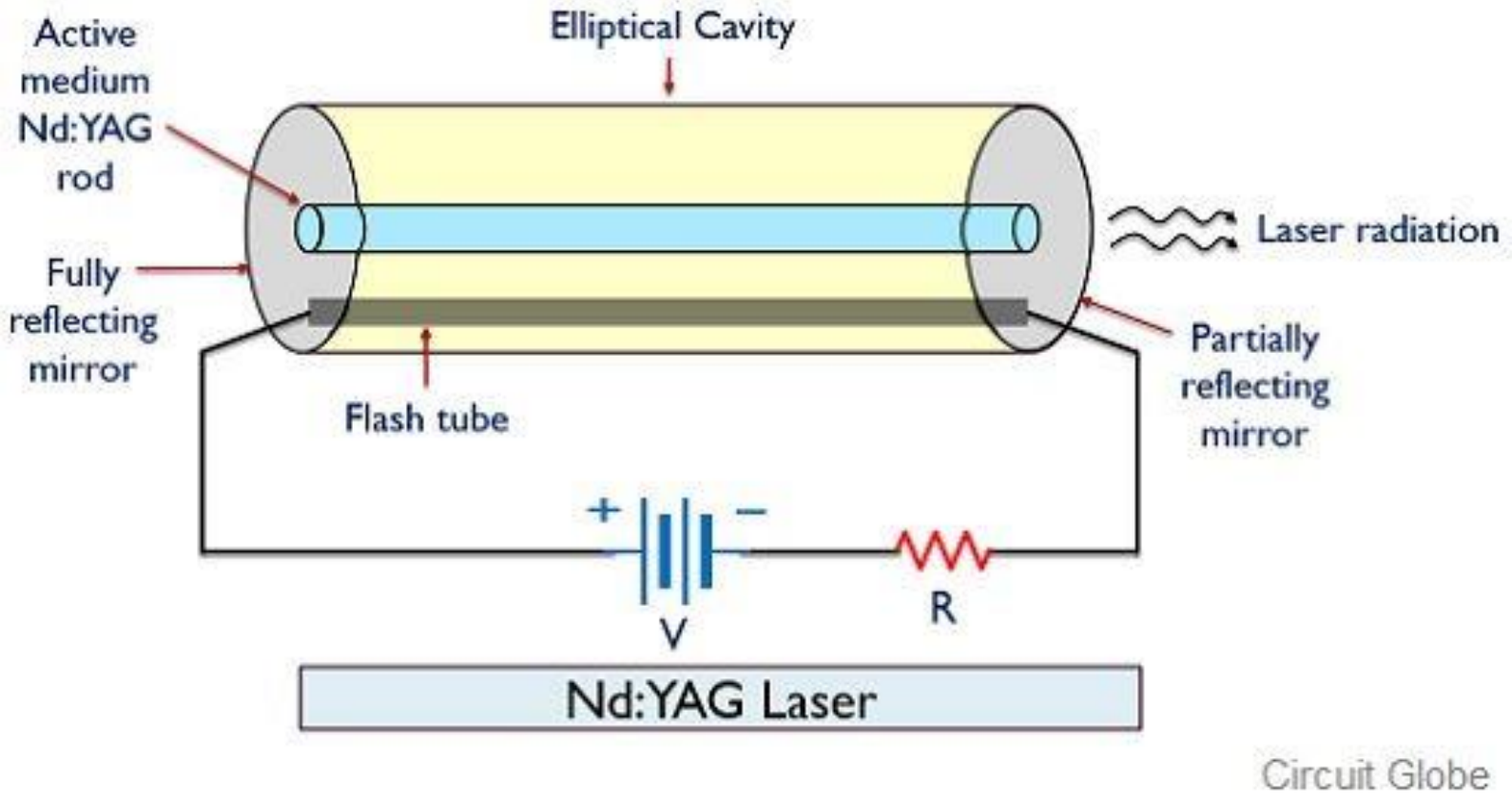
(c) Population Inversion for Neon :

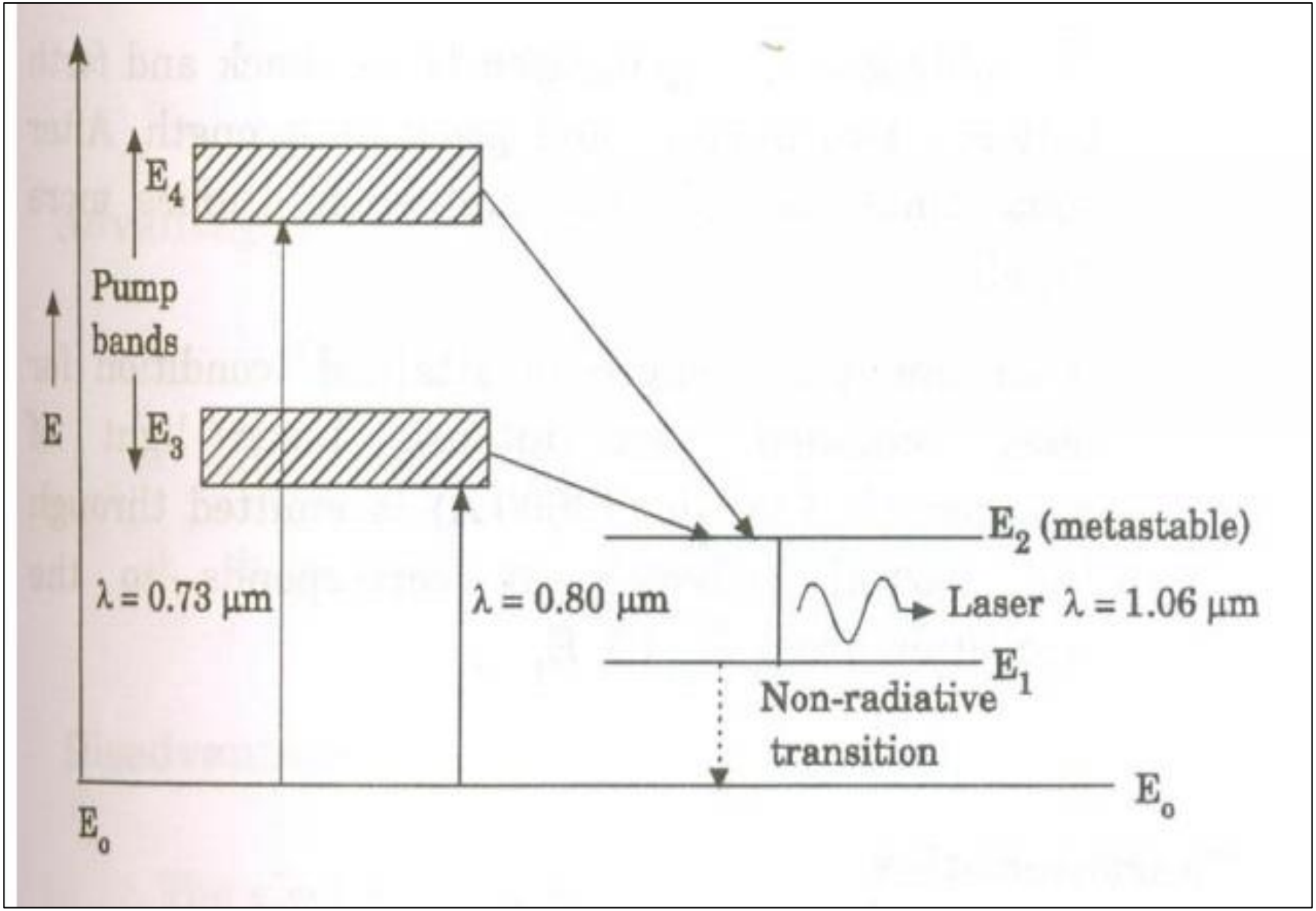
The transition between Ne_6 and Ne_3 produces the popular 6328 A° (632.8 nm) line of the He - Ne laser. Neon atoms de-excite through spontaneous emission from Ne_3 to Ne_2 (lifetime ~10^{-8} sec.). As this time is shorter than the lifetime of level Ne_6 (~10^{-7} sec.), steady-state population inversion can be achieved between Ne_6 and Ne_3. Level Ne_2 is metastable and thus tends to collect atoms. The atoms from this level fall back to the ground level mainly through collisions with the walls of the tube. As Ne_2 is metastable, it is possible for the atoms in this level to absorb the spontaneously emitted radiation in the $Ne_3 \rightarrow Ne_2$ transition to be re-excited to Ne_3. This tends to reduce the effect of inversion. It is for this reason that the gain in this laser transition is found to increase with decreasing tube diameter.

(d) Transition within Neon and Continuous Output :

The other two important wavelengths from the He - Ne laser correspond to the $Ne_4 \rightarrow Ne_3$ (1.15 μm) and $Ne_6 \rightarrow Ne_5$ (3.39 μm) transitions. The laser can be made to oscillate at 6328 A° by using optical elements (multilayer-coated mirrors) in the path. These lasers are continuous because the collision process maintains the energy states Ne_6 and Ne_4 at larger population densities than the lower states. This continued population inversion gives a continuous lasing action.

Nd-YAG LASER


Active medium Nd:YAG rod
Elliptical Cavity
Laser radiation
Fully reflecting mirror
Partially reflecting mirror
Flash tube
+
−
V
R
Nd:YAG Laser
Circuit Globe


E4
Pump
bands
E
E3
E2 (metastable)
λ = 0.73 μm
λ = 0.80 μm
Laser λ = 1.06 μm
E1
Non-radiative
transition
E0
E0

Nd YAG LASER

- ***Introduction*** – Nd: YAG laser is a neodymium-based laser. Nd stands for Neodymium (rare earth element) and YAG stands for Yttrium Aluminum Garnet (Y3Al5 O12) . It is a four-level solid state laser.

- ***Principle*** – The active medium Nd: YAG rod is optically pumped by Krypton flash tubes. The Neodymium ions (Nd3+) are raised to excited levels. During the transition from meta stable state to ground state, a laser beam of wavelength 1.064µm is emitted

Nd YAG LASER

Construction – The construction of the Nd: YAG laser is shown in the figure. A small amount of Yttrium ions (Y3+) is replaced by Neodymium (Nd3+) in the active element of Nd: YAG crystal. This active element is cut into a cylindrical rod. The ends of the cylindrical rod are highly polished, and they are made optically flat and parallel. This cylindrical rod (laser rod) and a pumping source (flash tube) are placed inside a highly (reflecting) elliptical reflector cavity. The optical resonator is formed by using two external reflecting mirrors. One mirror (M1) is 100% reflecting while the other mirror (M2) is partially reflecting.

Nd YAG LASER - *Working*

The figure shows the energy level diagram for the Nd: YAG laser. These energy levels are those of Neodymium (Nd3+) ions.

1. When the krypton flash lamp is switched on, by the absorption of light radiation of wavelength 0.73µm and 0.8µm, the Neodymium(Nd3+) atoms are raised from ground level E0 to upper levels E3and E4 (Pump bands).
2. The Neodymium ions atoms make a transition from these energy levels E2 by non-radiative transition. E2 is a metastable state.
3. The Neodymium ions are collected in the level E2, and the population inversion is achieved between E2 and E1.
4. An ion makes a spontaneous transition from E2 to E1, emitting a photon of energy hγ. This emitted photon will trigger a chain of stimulated photons between E2 and E1.
5. The photons thus generated travel back and forth between two mirrors and grow in strength. After some time, the photon number multiplies more rapidly.
6. After enough strength is attained (condition for laser being satisfied), an intense laser light of wavelength 1.06µm is emitted through the partial reflector. It corresponds to the transition from E2to E1

Nd YAG LASER – Characteristics

1. **Type:** It is a four-level solid-state laser.

2. **Active medium**: The active medium is Nd: YAG laser.

3. **Pumping method**: Optical pumping is employed for pumping action.

4. **Pumping source**: Xenon or Krypton flash tube is used as a pumping source.

5. **Optical resonator**: Two ends of Nd: YAG rod is polished with silver (one end is fully silvered and the other is partially silvered) and are used as optical resonators.

6. **Power output**: The power output is approximately 70 watts.
7. **Nature of output**: The nature of output is pulsed or continuous beam of light.

8. **Wavelength of the output**: The wavelength of the output beam is 1.06µm(infra-red)

Nd YAG LASER

- **Advantages:**

1. It has high energy output.
2. It has very high repetition rate operation
3. It is much easy to achieve population inversion.

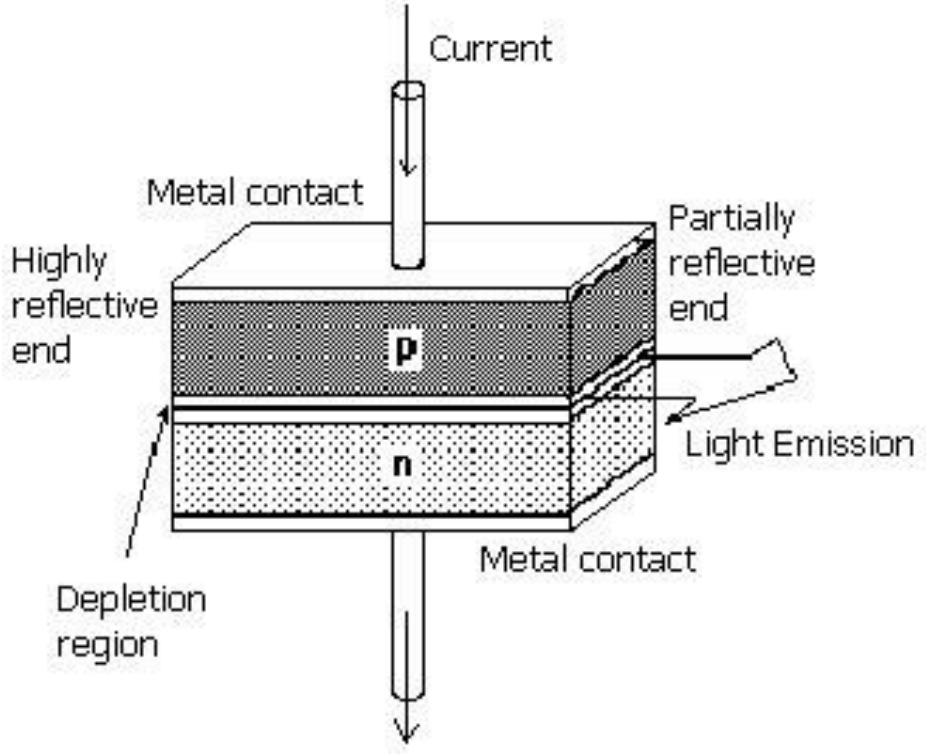
- **Disadvantages:**

1. The electron energy level structure of Nd^{3+} in YAG is complicated.

- **Applications:**

1. It finds many applications in range finders and illuminators.
2. It is widely used in engineering applications such as resistor, trimming scribing, micro machining operations as well as welding, drilling etc.
3. It finds many medical applications such as endoscopy, urology, neurosurgery, ENT, gynecology, dermatology, dental surgery and general surgery.

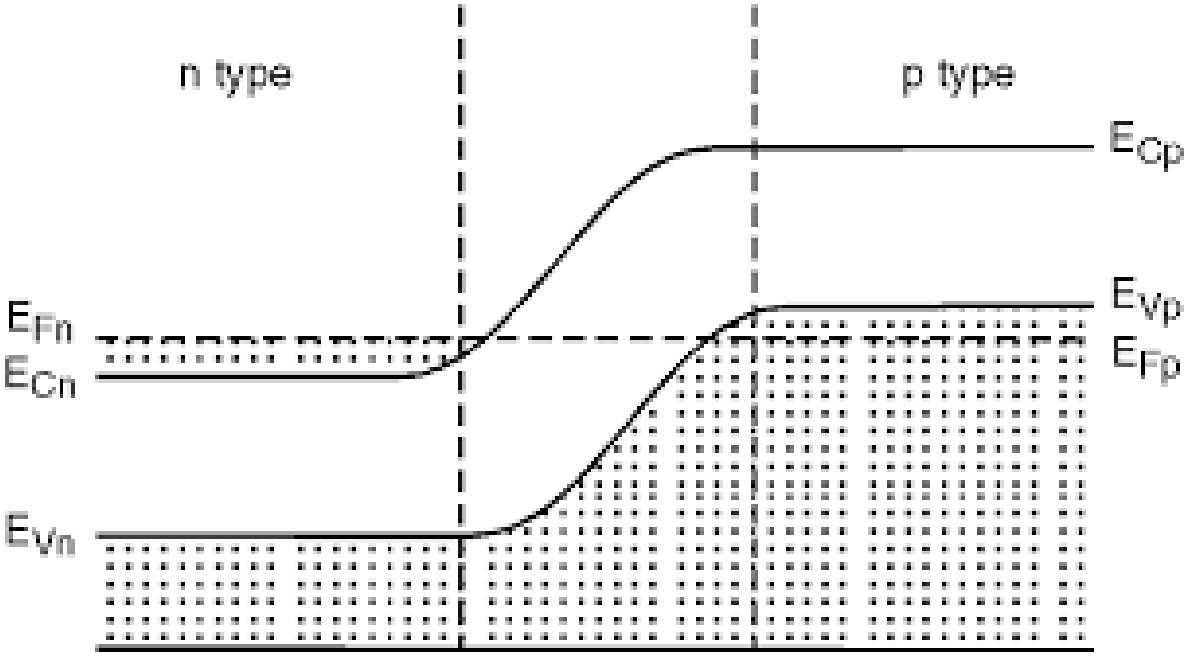
Semiconductor laser



A semiconductor diode laser is a p-n junction device that emits coherent light when forward biased. A p-n junction laser is also called **'injection laser'** as the charge carriers are injected into depletion region. The diode lasers are preferred over other lasers as they are compact, reliable and low cost.

- **Construction :** A laser diode is extremely small in size with sides of the order of 1 mm. The p-n junction of the diode lies in the horizontal plane. The top and bottom faces are metalized for making electrical contacts. Two plane mirrors are placed at front and back surface to form optical cavity. Sometimes the faces itself are polished to form optical cavity. The other two faces are roughened to prevent lasing action in that direction.

Semiconductor laser



Working :

- **(a) Under Equilibrium :** In semiconductor diode laser, the p-n semiconductor is heavily doped The Fig. shows the energy band of highly doped p-region and n-region. Due to very high doping on n-side, the donor level as well as a portion of conduction band is occupied by electrons. Hence, the Fermi level lies within the conduction band. Similarly, on the heavily doped p-side, holes exist in the valence band and hence the Fermi level lies in the valence band.

Semiconductor laser

- **(b) Forward Bias, Pumping and Recombination :** At low forward current, the electron-hole recombination cause spontaneous emission of photons and the junction acts as an LED. As the current is increased, the intensity of light increases. When the current reaches a threshold value, the carrier concentration in the depletion region reaches a very high value i.e. a large concentration of electrons in the conduction band and a large concentration of holes in the valence band. As a result, the higher energy levels in the depletion region are having high population density of electrons while the lower levels are vacant. This is a state of population inversion. The region where the population inversion is achieved is called as **active region**. The forward bias acts as the pumping agent.
- **(c) Built-Up of Laser :** Due to forward bias, the band gap is very small and therefore electrons in the conduction band can recombine with holes in the valence band. During this process of recombination of electrons and holes, electrons lose their energy emitting a photon of energy equal to band gap. Some of the photons emitted during the process of recombination are reflected back from the polished ends and stimulate the electrons initiating the laser process. The highly polished ends continuously reflect the light back and forth. The photons parallel to the axis of resonator will leak out as laser beam.

Semiconductor laser

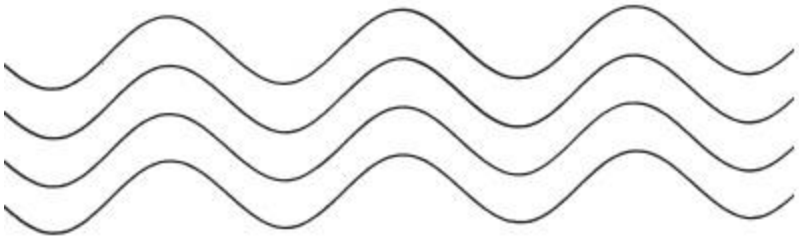
- The frequency of radiated photon is
  - $\nu = E_g/h$,
  - where E_g is band gap and h is Planck's constant. The wavelength of laser beam depends upon the material used, for GaAs laser wavelength is 9000 A° in IR region and GaAsP it is 6500 A° in visible region at room temperature.

- **Advantages of Semiconductor Laser :** Very less power required, simple, compact, highly efficient,
- **Disadvantages** - more divergent, less monochromatic and highly temperature sensitive compared to other types of laser.

Characteristics of LASER

- **Monochromaticity -** When the laser produces only one wavelength then it is fully monochromatic. This is impossible, in principle and in practice also. If 'Δλ' is the range of wavelengths included in a laser beam of wavelength l, then 'Δv' is the corresponding frequency. This frequency band 'Δv is called the **'line width'**. For white light consisting of all visible frequencies, the line width is ~10^{14} Hz, while for a good laser it is about 10^{2} Hz.

- **Coherence -** The light emerging from a laser is coherent both in space and time. The existence of finite bandwidth Δv means that the different frequencies present in laser can get out of phase width each other.



Coherent Light Waves

Properties of lasers

- **Directionality -** A laser has a high degree of directionality and can travel very large distances without deviation. For a typical laser, the divergence is about 10^{-3} radians. This means that the laser beam diverges by ~1 mm for every meter that it travels. The reason is that the active material is placed in a resonant cavity. The light is reflected back and forth in the cavity and light travelling parallel to the axis gets emitted as the laser beam. The light traveling in the other direction is reflected in the cavity.
- **Brightness -** A laser radiates light into a narrow beam and its energy is concentrated in a small region. This is because the divergence is very less. This results in extremely high intensity. Even a 1 W laser appears more intense than an ordinary 100 W lamp. This high concentration is used in many applications such as drilling, welding, and cutting.

Applications of lasers

- **1 Engineering / Industry**
- Laser can be focussed to a very high energy density into a small image (≈ 1 micron in diameter) with the help of suitable lenses. Due to the small size of the image and the control over the energy, lasers are used extensively for cutting, welding, and drilling circuits.
- **Drilling:** A laser beam is also used to drill holes of micron dimensions on printed circuit boards (PCBs). It is also used in resistance trimming in electric components industries. One can drill holes of a diameter of 10 mm through very hard substances like diamonds. YAG laser is found to be very useful in such applications.
- **Welding:** Lasers are used as a heat source in welding the joints of the metals. This type of precise welding is extremely important in micro-electronics in which thin films are used. Thermocouple wires can easily be welded with the help of a high-power laser beam.
- **Micromachining:** Lasers are used for machining a surface slowly and accurately to achieve an extraordinarily smooth finish.

Applications of lasers

- **Cutting :** Another important industrial application is metal or fabric cutting. A finely focused laser beam can cut thick and hard metal sheets with high precision and accuracy. It is also used in tailoring industries to cut thousands of layers of cloth at one instant.
- **Surveying :** Due to its intensity and directionality, laser is used in surveying. When tunnels are to be constructed, engineers use the laser beam as a reference, to check that it is being constructed along a straight line. Its most interesting use in surveying has been in measuring the distance from the earth to the moon. This distance was measured to an accuracy of 600 ft.
- A laser beam can determine precisely the distance, velocity and direction as well as the size and form of distant objects by means of the reflected signal as in radar. A Lidar (Laser radar), which sends out beams of laser light and detects echoes even from atmospheric layers has been developed.

Applications of lasers

- **2 Medicine**
- Bloodless cancer surgeries can be performed as the beam can be focused on a small area, so that only the harmful tissue can be destroyed without damaging the surrounding region.
- Laser has been successfully used in ophthalmology, in the treatment of detached retinas, in welding cornea, etc.
- Lasers have been built into or are devised to be attached to microscopes. As a high density energy is achieved, it can be used in micro-surgery, micro-burning, etc. Such a microscopic laser can concentrate millions of watts of power per square millimeter into a selected area.
- Laser microprobes can be used as dental drills giving an advantage of no heating, no anesthetic and no pain to the patient. They have also been successfully used for localized treatment of skin growths. A large amount of energy can be transmitted through the skin to interact with deeper different biological materials or structures which are damaged.

Applications of lasers

Computer –

CD Writer Devices : In the compact disk (CD), the pick-device or writing-device does not come into contact with the disc surface, therefore there is no degradation of the data while recording or reproduction. The recording of information on CD is done in binary form, i.e. in 1s and 0s with a laser beam and reading of the recorded information is also done by a low-powered laser beam..

A laser beam is used to write data on the disk in digital form as tiny **'pits'** and **'lands'** (**'lands'** are reflective sections between pits). The laser beam starts from the innermost spiral of the disk and travels outwards as the disk rotates. The data to be recorded on the CD is given to a diode laser and the intensity of laser beam is controlled accordingly. The laser beams is focused on the disk surface which is coated on a special material. Depending on the material used, the laser beam either burns tiny holes or creates microscopic bubble in the place of 1s.

While reading a laser beam of lower intensity is focused on the track and the reflected beam is sensed by a photodiode. As the holes or bubbles scatter the laser beam, very little light is reflected from these positions thereby detecting 1s.

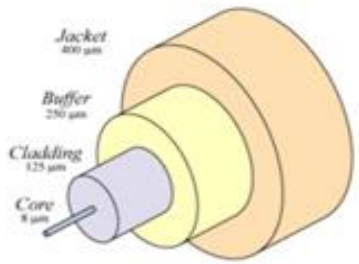
Laser Printers : In case of laser printers, the image (data to be printed) is produced on the photo-sensitive drum with a laser beam.

Principle and structure of optical fibre

- **Principle of Light Transmission :** The principle of light transmission through optical fibre is total internal reflection. For total internal reflection to take place at the fibre wall, the following conditions should be satisfied :
  - The refractive index of the core material (μ_1) must be greater than that of the cladding (μ_2).
  - At the core-cladding interface, the angle of incidence Θ must be greater than the critical angle, where $\theta_c = \sin^{-1}\left(\frac{\mu_2}{\mu_1}\right)$
- When light is incident on core of the fibre optics, it will be refracted and will travel in the core. After some time, it will strike one of the core-cladding interface say upper surface. If the angle of incidence is greater than critical angle, it will be totally reflected and remain in the core. Now, the reflected light will travel to the lower surface. It is then incident on the lower surface where the same process is repeated, and light gets transmitted from one end to the other end as shown in Fig.

BASIC STRUCTURE

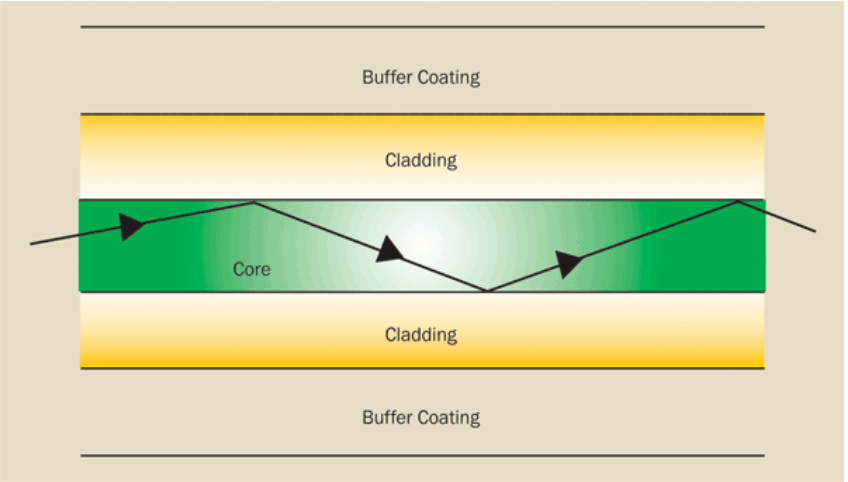
- An optical fiber contains three layers:



1.Core :It carries the light signals.

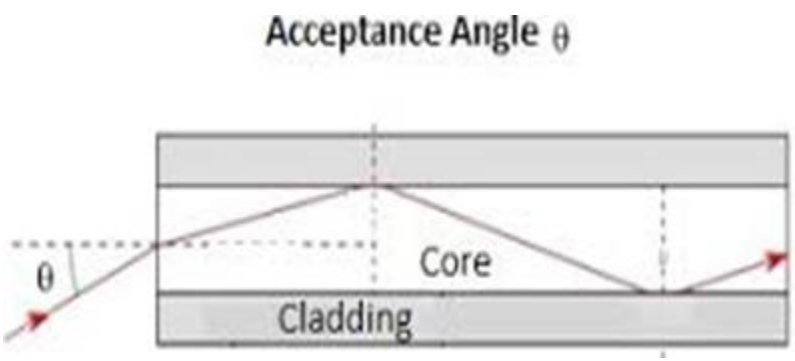
2.Cladding:It keeps the light in the core.

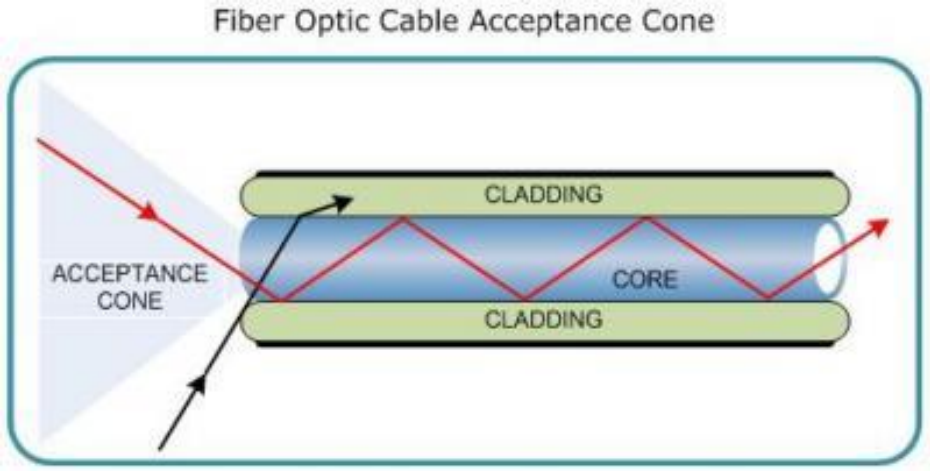
3.Coating: It protects the cladding from damage.



Acceptance angle, Acceptance cone

- **Acceptance angle-** is a maximum angle that a light ray can have relative to the axis of the fibre and propagate down the fibre (Θ_0). The maximum angle of incidence that ray can make which will result in total internal reflection. Thus, only those rays that are incident on the face of the fibre making angles less than Θ_0 will undergo repeated total internal reflections and reach other end of fibre.
- **Larger acceptance angles make it easier to launch the light into the fibre. But beyond Θ_0 light refracts through the cladding and the corresponding optical energy is lost.**
- **Acceptance cone** – in 3D, the light ray contained within the cone having full angle $2\Theta_0$ are accepted and transmitted along the fibre. Therefore, the cone is called as acceptance cone. If all possible directions of the acceptance angle is considered, we get a cone known as acceptance cone.





Numerical aperture

- The main function of the optical fibre is to accept and transmit as much light as possible from the source. The light gathering ability of the optical fibre depends upon core size and numerical aperture.
- The acceptance angle and the fractional refractive index (fractional difference between the R.I of core and cladding) change determines the numerical aperture of the fibre.
- Numerical aperture –Light gathering power of the fibre. sine of the acceptance of angle.

$$NA = \sqrt{\mu_1^2 - \mu_2^2}$$

- It describes the cone of light accepted into the fibre or exciting the fibre.
- NA Value ranges from 0.13 to 0.50. **Large NA indicates that fibre will accept large amount of light.**
- **Also, only those rays which are refracted into the cable at angles $\Theta_r < \Theta_c$ will propagate in the optical fibre.**

Types of Optical Fibers

- **The types of optical fibers depend on the refractive index, materials used, and mode of propagation of light.**

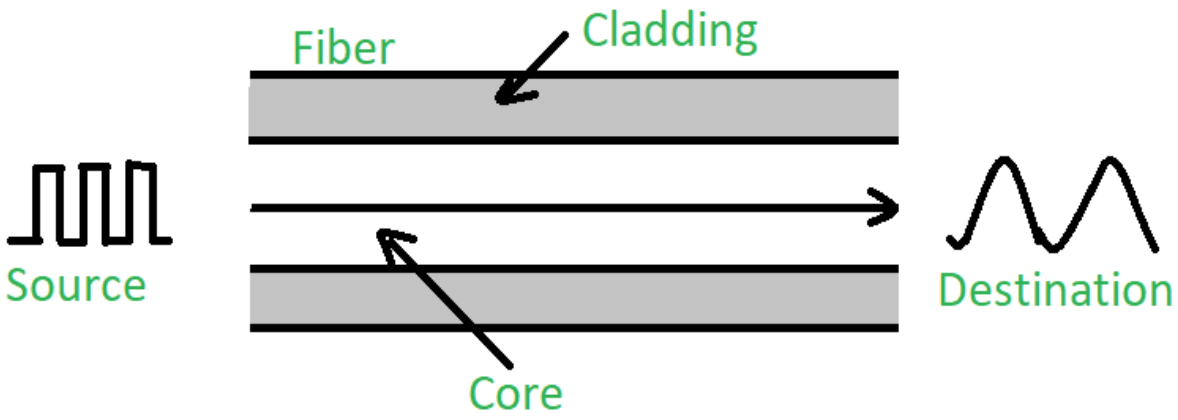
- **The classification based on the materials used is as follows:**
- **Plastic Optical Fibers:** The polymethylmethacrylate is used as a core material for the transmission of the light.
- **Glass Fibers:** It consists of extremely fine glass fibers.

- **The classification based on the refractive index is as follows:**
- **Step Index Fibers:** It consists of a core surrounded by the cladding, which has a single uniform index of refraction.
- **Graded Index Fibers:** The refractive index of the optical fiber decreases as the radial distance from the fiber axis increases.

- **The classification based on the mode of propagation of light is as follows:**
- **Single-Mode Fibers:** These fibers are used for long-distance transmission of signals.
- **Multimode Fibers:** These fibers are used for short-distance transmission of signals.
- The mode of propagation and refractive index of the core is used to form four combination types of optic fibers as follows:
- Step index-single mode fibers
- Graded index-Single mode fibers
- Step index-Multimode fibers
- Graded index-Multimode fibers

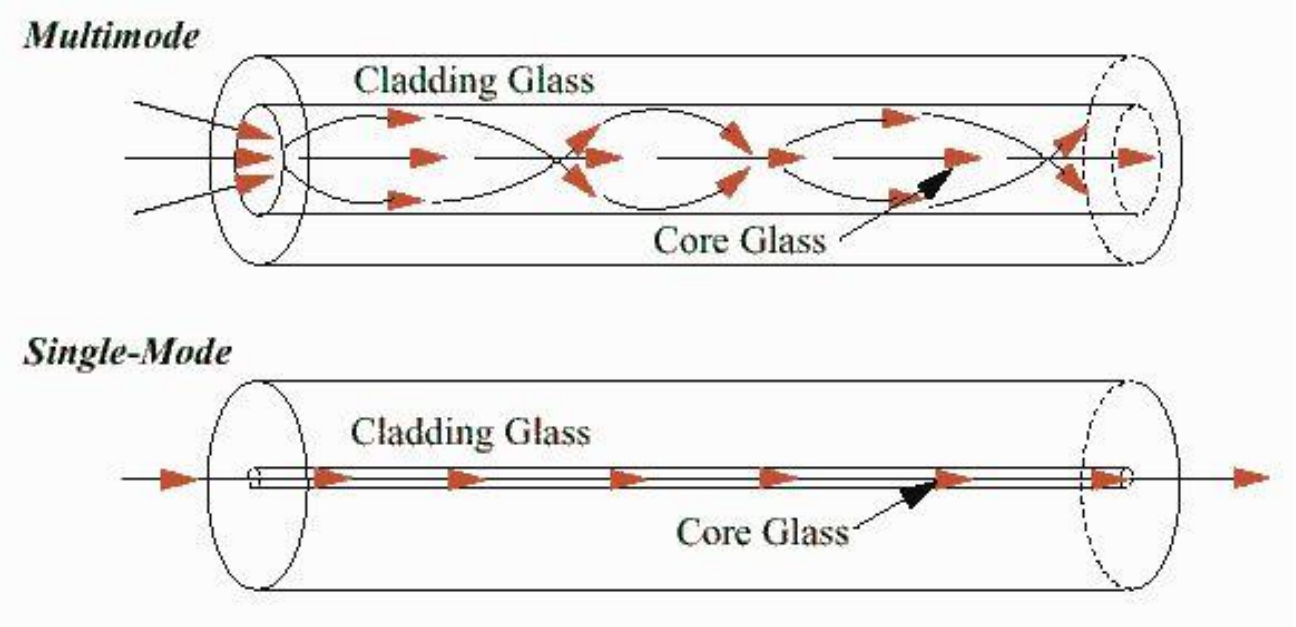
Types of Optical Fibers

- **Generally optical fiber is classified into two categories based on: *the number of modes, and the refractive index*.**
- **On the basis of the Number of Modes**
  It is classified into 2 types
- **Single-mode fiber:**
  In single-mode fiber, only one type of ray of light can propagate through the fiber. This type of fiber has a small core diameter (5um) and high cladding diameter (70um) and the difference between the refractive index of core and cladding is very small. There is no dispersion i.e. no degradation of the signal during traveling through the fiber. The light is passed through it through a laser diode.



Types of Optical Fibers

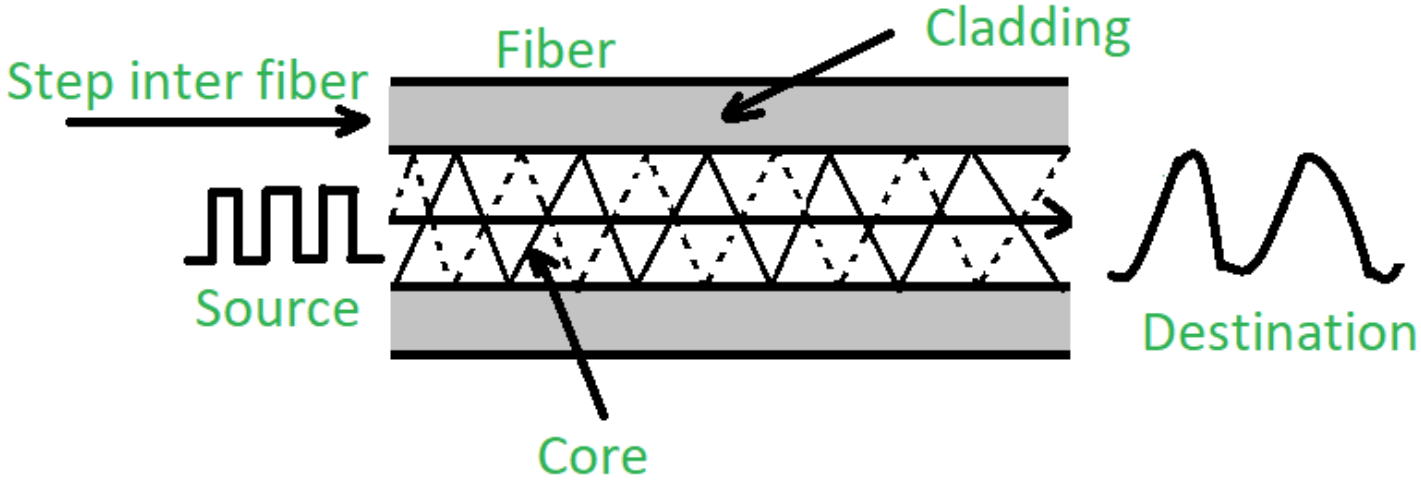
- **(b). Multi-mode fiber:** Multimode fiber allows a large number of modes for the light rays traveling through it. The core diameter is generally (40um) and that of cladding is (70um). The relative refractive index difference is also greater than single-mode fiber. There is signal degradation due to multimode dispersion. It is not suitable for long-distance communication due to the large dispersion and attenuation of the signal. There are two categories on the basis of Multi-mode fiber i.e. Step **Index Fiber** and **Graded Index Fiber. Basically,** these are categories under the types of optical fiber based on Refractive Index



Types of Optical Fibers

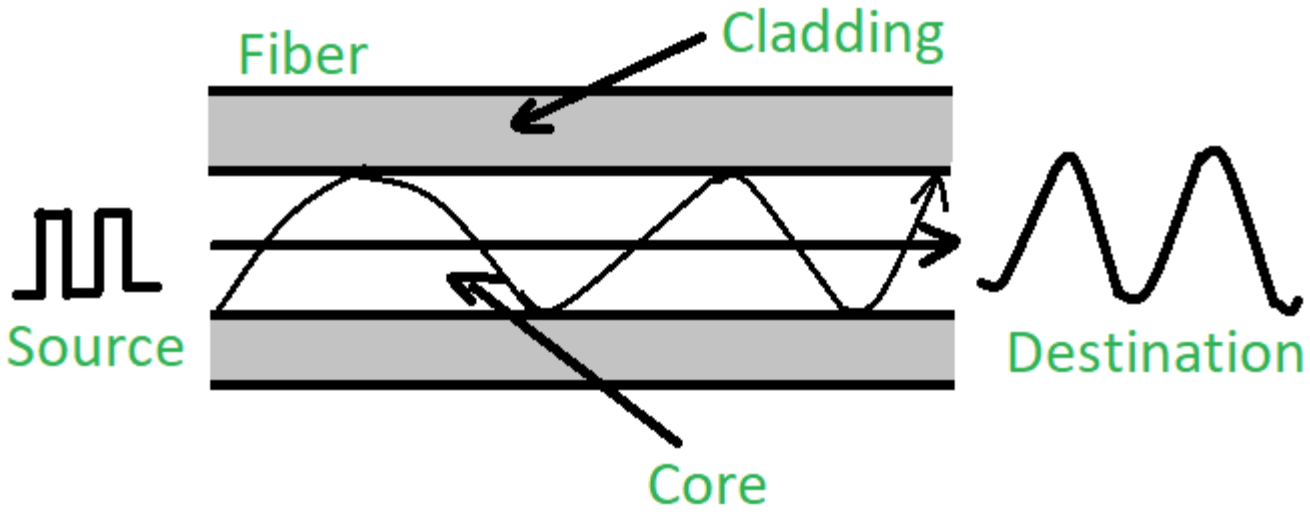
- **Based on Refractive Index:**
  It is also classified into 2 types:

- **Step-index optical fiber :-**
  The refractive index of core is constant. The refractive index of the cladding is also constant. The rays of light propagate through it in the form of meridional rays which cross the fiber axis during every reflection at the core-cladding boundary.

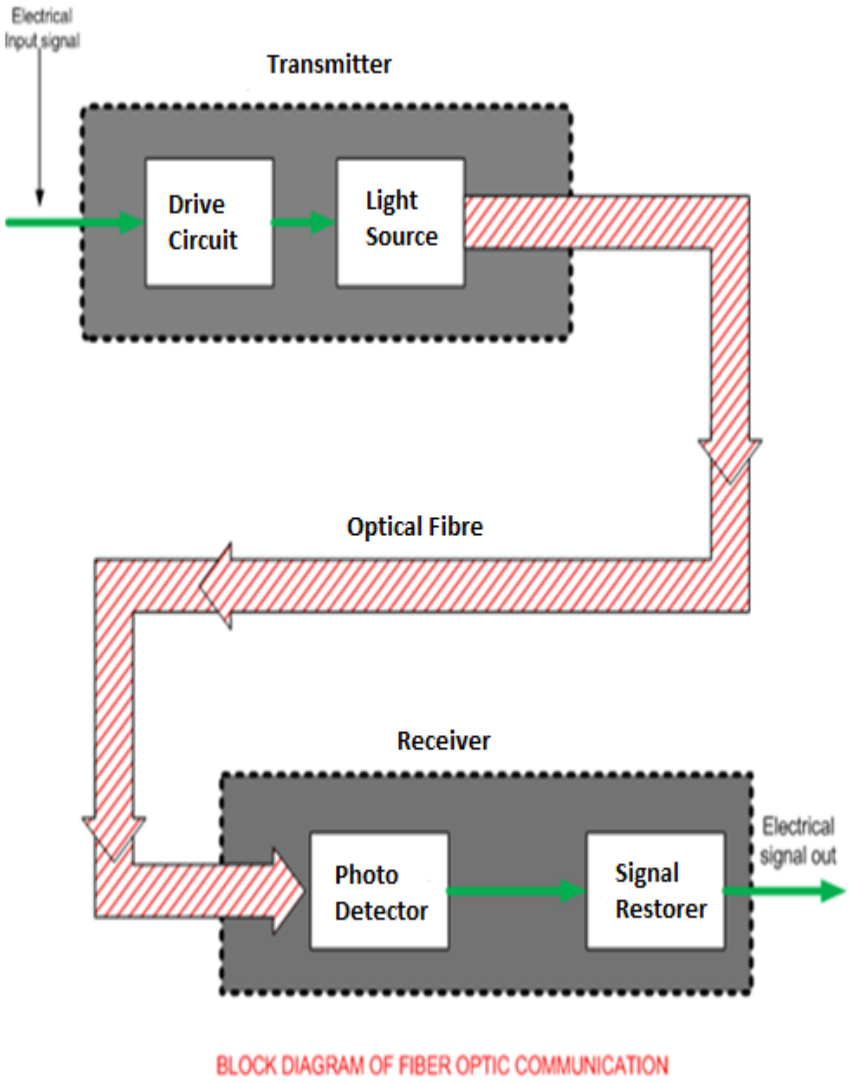


Types of Optical Fibers

- **Graded index optical fiber:**
  In this type of fiber, the core has a non-uniform refractive index that gradually decreases from the center towards the core-cladding interface. The cladding has a uniform refractive index. The light rays propagate through it in the form of skew rays or helical rays. It will not cross the fiber axis at any time.



Applications of optic fibre



BLOCK DIAGRAM OF FIBER OPTIC COMMUNICATION

- **Communication Kit :** One of the most important areas of application of fibre optics is in telecommunication. The communication kit consists of a transmitter, optical fibre and the receiver. The block diagram is as shown in the Fig.

  The transmitter consists of a light source, a laser diode, supported by necessary Drive circuit. The transducer converts nonelectrical signals into electrical signals and fed to light source. Varying intensity – analog modulation. Switching ON (1) OFF(0) – digital modulation. The light beam from the source is connected to the fibre, through optical connections. The carrier and signal frequency propagate through the fibre. The receiver is programmed to recognize digital pattern of signal. At the other end, it is detected with the help of a photo-detector – converts light signal to electrical signals – amplified – decoded to obtain original message.

- Transducer converts it into A/V form.

Formulas

- Critical angle - $\sin\theta_c = \frac{\mu_2}{\mu_1}$
- Acceptance angle – $\theta_0 = \sin^{-1}\sqrt{\mu_1^{\,2} - \mu_2^{\,2}}$
- Angle of acceptance cone – $\mathbf{2\Theta_0}$
- Numerical Aperture - $NA = \sqrt{\mu_1^{\,2} - \mu_2^{\,2}}$

Numericals

- In an optical fibre, core material has refractive index of 1.6 and R.I clad material is 1.3. what is the value of critical angle? Also calculate the value of angle of acceptance cone.

- Fibre cable has ac accept angle of 30^0 and a core index of refraction 0f 1.4. Calculate the R.I of cladding.

- Calculate the numerical aperture and value of acceptance angle of an optical fibre from the following data,

  $\mu_{1(core)}=1.55$ $\mu_{2(cladding)}=1.50$